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Towards generalizable face forgery detection via mitigating spurious correlation

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ABSTRACT

The continuous advancement of face forgery techniques has caused a series of trust crises, posing a significant menace to information security and personal privacy. In response, deep learning is being employed to develop effective detection methods to identify deepfake images and videos. Currently, most detection methods generally achieve satisfactory performance in intra-domain detection. However, their effectiveness remains uncertain when detecting unknown forgeries and cross-domain scenarios. From a causal learning perspective, our analysis reveals that the failure of detectors in cross-domain detection primarily arises from spurious correlations between irrelevant features and category labels (real and fake). In this study, we propose the Feature Independence Constrainer (FIC) to alleviate spurious correlations and enhance the generalization for detection method. Specifically, FIC maps features to a Reproducing Kernel Hilbert Space and calculates the covariance matrix among the kernel-mapped features. Dynamic weight parameters are incorporated into the covariance matrix for iterative optimization. Consequently, statistical correlations between features are eliminated, ensuring independence between each pair of features and mitigating the impact of spurious correlations. Additionally, we introduce fine-grained high-frequency components to force the detection model to learn comprehensive forgery-related artifacts while avoiding spurious correlations between irrelevant features and labels. Furthermore, a Feature Alignment Module (FAM) is designed to explore higher-order dependencies between the spatial and frequency domains, allowing for the extraction of richer potential forgery traces. Quantitative and qualitative experiments demonstrate that the proposed method achieves competitive performance across multiple face forgery benchmark datasets, outperforming state-of-the-art methods.

1. Introduction

The remarkable advances in generative models have led to significant progress in face forgery techniques (Liu, Li, Deng, & Sun, 2022). However, the widespread abuse of numerous counterfeit images and videos has raised concerns among the public and society, posing serious challenges to privacy protection and information security. In response, researchers have focused on developing effective detectors (Qiao, Xie, Chen, Retraint, & Luo, 2024; Xia et al., 2024) to identify forgeries and prevent the malicious dissemination of manipulated faces. Due to advancements in generation techniques and the continuously improving quality of forgeries, most detectors face a constant challenge in keeping pace with the sophistication of fraudulent methods. The detection approaches and forgery techniques are evolving in a game-like manner. Under these circumstances, there is an urgent need to develop consistently effective and generalized detectors to counter the threat posed by deepfake technology.

The key factor in distinguishing forged face images from real ones lies in exposing forgery traces. Through extensive investigation and research, we have categorized the existing deepfake detection methods into three categories: (1) Detection methods that rely on data enhancement or auxiliary information. This type of method (Guan et al., 2022; Guo, Yang, Chen and Sun, 2023; Jia et al., 2022; Li, Bao, Zhang et al., 2020; Liu, Yu, Li, Zhao and Guo, 2023; Qian, Yin, Sheng, Chen, & Shao, 2020; Sun, Han, Hua, Ruan, & Jia, 2021; Wang & Deng, 2021; Yang, Zhou et al., 2023; Zhu et al., 2023) (see Fig. 1(a)) involves enhancing the training data, such as applying rotation and Gaussian blurring to improve the model's generalization ability. It also includes incorporating auxiliary information like edges and DCT transform coefficients to enable the detection network to capture potential forgery

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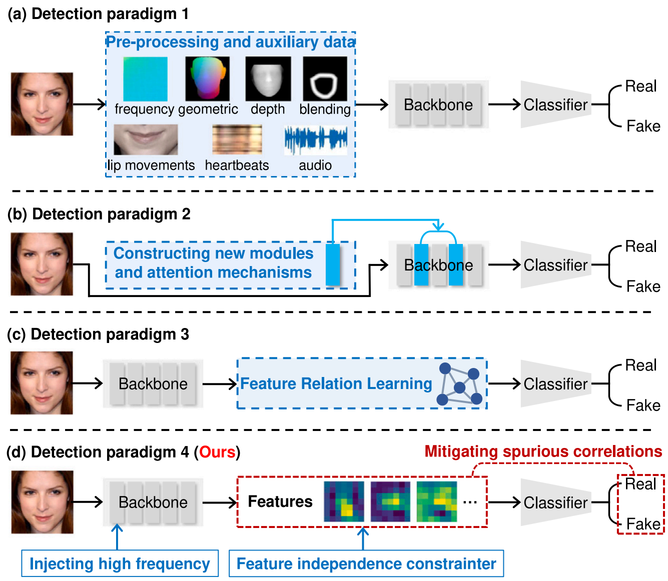

Fig. 1. Comparison of three mainstream face forgery detection paradigms with our method. (a) Detection methods that rely on data enhancement or auxiliary information. (b) Detection methods utilizing tailored network architectures. (c) Detection methods based on feature enhancement. Our approach (d) focuses on mitigating spurious correlations between irrelevant features and labels.

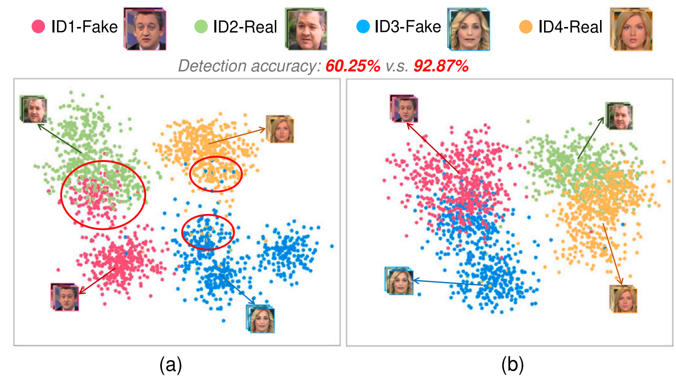

Fig. 2. We construct a unbalanced dataset to validate spurious correlations. The unbalanced dataset consists of 2000 fake male images and 2000 real female images. We train on the ID imbalance dataset with Xception and test on 500 face images each of ID1-Fake, ID2-Real, ID3-Fake, and ID4-Real. ID1-Fake and ID2-Real (ID3-Fake and ID4-Real) are semantically similar, age-, skin-color-, and hairstyle-similar male (female) face images. (a) We visualize the features learned by Xception using t-SNE. (b) Feature clustering results for Xception with the Feature Independence Constrainer (FIC).

features that are difficult to detect. (2) Detection methods utilizing tailored network architectures. This type of method (Bai, Liu, Zhang, Li, & Hu, 2023; Gu et al., 2021; Gu, Yao, Chen, Ding, & Ma, 2022; Guo, Jia et al., 2023; Lu et al., 2023; Song et al., 2022; Sun, Liu et al., 2022; Wang, Yu, Chen, Hu, & Peng, 2023; Zhao et al., 2021) (see Fig. 1(b)) constructs new network architectures or enhances existing classical networks by introducing attention mechanisms that allow the model to learn local high-response forgery features. (3) Detection methods based on feature enhancement. This type of method (Cao et al., 2022; Guo, Zhen and Yan, 2023; Lin et al., 2023; Liu, Xie, Wang and Zha, 2023; Sun, Yao et al., 2022; Woo et al., 2022; Yang, Liang, Xu, Zhang and He, 2023; Yu, Li, Li, Zhou, & Lu, 2023) (see Fig. 1(c)) typically utilizes classical networks as feature extractors and introduces mechanisms like contrastive learning to understand the correlation between different features, thereby increasing the distance between distinct features and discovering more discriminative forgery features. In general, these methods exhibit remarkable success in intra-domain detection. However, due to the detection network's tendency to learn the inherent distribution patterns within datasets and the semantic information of images (e.g., facial content and identity), it is difficult for these methods to maintain their effectiveness when confronted with unknown forgery types or new datasets.

The inherent distribution patterns within dataset and semantic information of images (we call irrelevant features, which refer to the features that are not related to the forgery traces) usually do not exactly match the forgery features (relevant features). For example, for local forgery, in most cases, the forgery region does not exactly tailor the semantic content. Thus, the decision logic of the detection model would be misguided (Dong et al., 2023; Yan, Zhang, Fan, & Wu, 2023). From the perspective of causal learning (Guo, Zhang, Feng, & Chen, 2021; Tian, Wang, Chen, & Kwong, 2024), when the data distribution of the testing set deviates from that of the training set, the features extracted by the network will deviate from the forgery features, leading to inaccurate decisions. In other words, the model's performance degradation on unseen datasets can be attributed to spurious correlations between irrelevant features and category labels (real and fake) (Wang, Qi, Guo, Shi, & Gao, 2024). This spurious correlation is essentially caused by the statistical association between irrelevant and relevant features. To verify the spurious correlation in face forgery detection, we construct an unbalanced dataset (the training set contains 2000 forged male face images and 2000 authentic female face images) for experiments. We train the Xception (Chollet, 2017) network on the imbalance dataset, and use t-SNE (Van der Maaten & Hinton, 2008) to visualize the high-dimensional features extracted from the last layer of Xception, the results are shown in Fig. 2(a). As can be seen from Fig. 2(a), Xception primarily relies on identity information (such as gender, hairstyle, and skin color) for classification, resulting in the detection accuracy for real and fake is only 60.25%. This suggests the presence of a spurious correlation between identity information (irrelevant features) and the label, which significantly impacts the decision-making process of the model. Moreover, an excessive reliance on this spurious correlation by the detection model severely hampers its generalization performance.

Motivated by the above observation, in this study, our focus is on mitigating the spurious correlations between irrelevant features and category labels by attenuating the statistical correlation between relevant and irrelevant features. However, due to the complex nonlinear dependencies and high correlations among features of face images, it is difficult to completely eliminate the statistical associations between these features. To address this issue, we propose the Feature Independence Constrainer (FIC) to eliminate the statistical associations between each pair of features to mitigate spurious correlations. Initially, we map the features to the Reproducing Kernel Hilbert Space via the kernel equation in FIC. Subsequently, the covariance matrix between the kernel-mapped features is computed. The statistical associations between features are thus converted into an explicit computation of the covariance matrix of the kernel-mapped features. To ensure independence between feature pairs, FIC incorporates a set of dynamic weight parameters for iterative optimization of the covariance matrix. Finally, by optimizing model and weight parameters, FIC minimizes statistical associations between relevant and irrelevant features. We use visualizations to demonstrate that FIC is able to mitigate the spurious correlation between irrelevant features and category labels, the results are shown in Fig. 2(b) (the detection accuracy is improved to 92.87%). Additionally, we incorporate fine-grained high-frequency components into the backbone network. The detection model is capable of acquiring a deeper understanding of forgery-related artifact concepts while mitigating the risk of learning spurious correlations between irrelevant features and labels to a certain extent. Moreover, we design the Feature Alignment Module (FAM) to explore higher-order correlations between spatial and frequency domains in order to extract more comprehensive potential forgery traces. Thanks to these meticulous designs, our

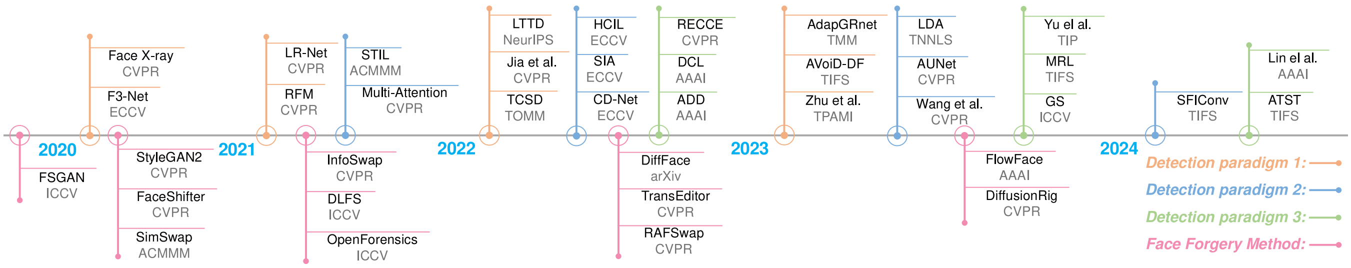

Fig. 3. Part of the superior works on detection methods (Face X-ray (Li, Bao, Zhang et al., 2020), F3-Net (Qian et al., 2020), LR-Net (Sun et al., 2021), RFM (Wang & Deng, 2021), TCSD (Liu, Yu et al., 2023), Jia et al. (Jia et al., 2022), LTTD (Guan et al., 2022), AdapGRnet (Guo, Yang et al., 2023), AVoiD-DF (Yang, Zhou et al., 2023), Zhu et al. (Zhu et al., 2023), STIL (Gu et al., 2021), Multi-Attention (Zhao et al., 2021), CD-Net (Song et al., 2022), SIA-Net (Sun, Liu et al., 2022), HCIL (Gu et al., 2022), LDA (Lu et al., 2023), AUNet (Bai et al., 2023), Wang et al. (Wang et al., 2023), SFIConv (Guo, Jia et al., 2023), ADD (Woo et al., 2022), DCL (Sun, Yao et al., 2022), RECCE (Cao et al., 2022), Yu et al. (Yu et al., 2023), MRL (Yang, Liang et al., 2023), GS (Guo, Zhen et al., 2023), Lin et al. (Lin et al., 2023), ATST (Liu, Xie et al., 2023), Liang et al. (Liang et al., 2023), Df-udetector (Ke & Wang, 2023)) and face forgery techniques (FSGAN (Nirkin, Keller, & Hassner, 2019), RAFSwap (Xu, Zhang et al., 2022), FlowFace (Zeng et al., 2023), DiffFace (Kim et al., 2022), TransEditor (Xu, Yin et al., 2022), DiffusionRig (Ding et al., 2023), StyleGAN2 (Karras et al., 2020), FaceShifter (Li, Bao, Yang, Chen and Wen, 2020), SimSwap (Chen, Chen, Ni, & Ge, 2020), InfoSwap (Gao, Huang, Fu, Li, & He, 2021), DLFS (He et al., 2021), OpenForensics (Le, Nguyen, Yamagishi, & Echizen, 2021)) in recent years.

approach achieves competitive performance on various face forgery detection benchmark datasets.

In summary, our contributions are summarized as follows:

- From the perspective of causal learning, we identify that the primary obstacle to model cross-domain detection is attributed to spurious correlations between irrelevant features and category labels.
- We propose the Feature Independence Constrainer (FIC) to mitigate spurious correlation by eliminating the statistical correlation between each pair of features. Ablation experiments demonstrate that FIC assists in improving the generalization performance of model.
- We design the Feature Alignment Module (FAM) to exploit higher-order dependencies between the spatial and frequency domains to mine richer potential forgery clues. FAM not only improves the detection accuracy and cross-domain capability of the model, but also enhances robustness of the model.

2. Related work

Due to the remarkable advancements in generative models, face forgery techniques have made significant strides. To prevent their illicit use, researchers have extensively explored forgery detection. In this section, we present the prevailing face forgery techniques and recent detection methods (depicted in Fig. 3). Additionally, we introduce the issue of spurious correlations in tasks related to natural language processing and computer vision.

2.1. Face forgery techniques

The manipulation of facial images primarily encompasses the forgery of face identity and face attributes. Face identity forgery typically involves transferring and exchanging source faces onto target faces. The remarkable success of generative adversarial networks (GANs) (Hu, Wang, & Zhang, 2023; Li et al., 2023) in computer vision has spurred extensive exploration into face-swapping models. FSGAN (Nirkin et al., 2019) comprises three generators and a semantic segmentation network, enabling the swapping of faces between any two subject-irrelevant images. RAFSwap (Xu, Zhang et al., 2022) achieves high-resolution face generation with consistent identities through a local–global approach. FlowFace (Zeng et al., 2023) is a shape-aware, semantic flow-guided framework that facilitates more realistic face swapping by transferring both facial features and contours to the target face. Recently, diffusion models have achieved significant success in generative tasks, with DiffFace being the first framework to utilize diffusion models for face swapping. Compared to GANs, DiffFace (Kim et al., 2022) offers superior advantages in terms of training stability, controllability, and fidelity. Face attribute forgery generally involves modifying expressions, hairstyles, ages, etc. TransEditor (Xu, Yin et al., 2022) introduces a novel approach to image editing and inverse extrapolation, employing a two-space framework. Additionally, it incorporates a Transformer-based cross-space interaction mechanism that facilitates precise control over facial attribute editing. DiffusionRig (Ding et al., 2023) leverages diffusion models to enable face attribute editing by learning both large-scale facial prior knowledge and individual-specific prior knowledge. However, the widespread adoption of these face manipulation technologies presents both advantages and disadvantages. While they bring convenience to digital entertainment and virtual reality applications, their misuse can lead to disinformation, privacy invasion, and fraud risks.

2.2. Face forgery detection methods

To combat the risks associated with face forgery, researchers have made significant advancements in developing superior forgery detection algorithms in recent years. In our investigation, we have categorized these detection methods into three paradigms. Detection Paradigm 1: F3-Net (Qian et al., 2020) incorporates two complementary frequency-aware cues into a collaborative learning framework consisting of two streams to effectively extract forgery features. TCSD (Liu, Yu et al., 2023) employs a depth estimator to perceive the depth information of the face image and combines it with the inconsistency between foreground and background information to capture forged patterns. Zhu et al. (2023) proposed a 3D decomposition-based detection method that decomposes face images into 3D shapes, illumination, and textures to emphasize hidden artifact details. Face X-ray (Li, Bao, Zhang et al., 2020) detects blending artifacts by constructing auxiliary data on blending boundaries within forged images. Additionally, there is auxiliary physiological cue information available to assist the detection model in identifying inconsistent features. Detection Paradigm 2: Multi-Attention (Zhao et al., 2021) comprises an attention module, a texture enhancement block, and an attention pooling layer that employ multi-attention mechanisms for capturing tampered features from multiple regions of the face. Wang et al. (2023) designed an attention graph learning module and a dynamic graph spatial-frequency feature fusion network to explore higher-order relationships among spatial-frequency features. Detection Paradigm 3: DCL (Sun, Yao et al., 2022) constructs positive-negative sample pairs using extracted features and combines inter-instance contrastive learning and intra-instance contrastive learning at different levels of granularity to acquire generalized feature representations. ATSC (Liu, Xie et al., 2023) adaptively mines intrinsically correlated forgery cues in both spatial and frequency domains to learn more generalized forgery features.

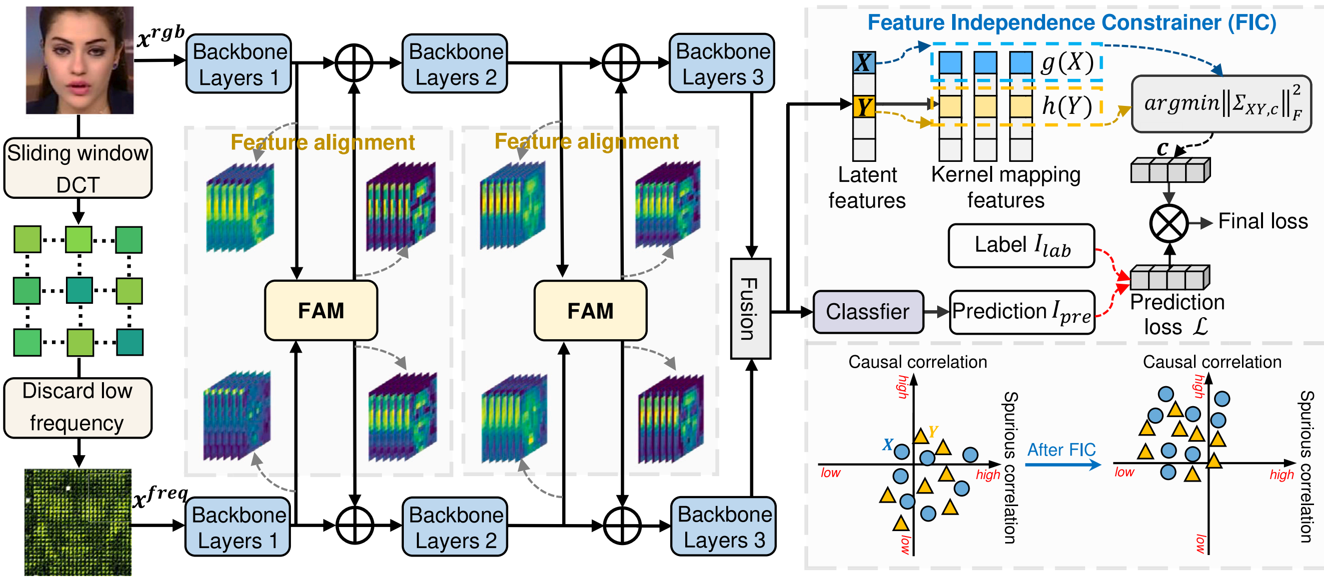

Fig. 4. Overview of the proposed method. The original RGB face image x^{rgb} is passed through the sliding window DCT and drop the low frequency to yield fine-grained high-frequency components x^{freq}. Xception is used as a backbone network to extract the spatial and frequency domain features and utilize the Feature Alignment Module (FAM) to align and fuse them at both shallow (block3) and deep (block11) layers. Feature Independence Constrainer (FIC) eliminates the statistical dependency of fused features to mitigate spurious correlation. The final Fusion block consists of depthwise convolution, batch normalization, ReLU activation function, and global average pooling.

2.3. Spurious correlation

Spurious correlation poses a common challenge in the domains of natural language processing and computer vision. Dou et al. (2022) investigated how natural language understanding models often rely on spurious correlations to achieve high performance within specific categories, while struggling with efficiency across different domains. To address this issue, they proposed feature de-correlation and feature purification techniques to compel the model to learn more task-relevant features. AutoACER (Kumar, Deshpande, & Sharma, 2024) automatically identifies spurious attributes by estimating their causal effect on labels and then employs a regularization objective to reduce the classifier's reliance on these attributes. Zhang et al. (2021) argue that mitigating distributional changes between training and testing data is crucial for enhancing model generalization capabilities. Therefore, they propose training sample weighting to eliminate correlations between features and alleviate spurious correlations. Huang et al. (2024), in the context of reinforcement learning, suggest that establishing genuine correlations between changing features and decisions helps agents accurately understand how features impact decision-making processes. They introduce saliency-guided feature de-correlation as an approach to eliminate spurious correlations. In face forgery detection, the presence of spurious correlations between irrelevant features and decision labels significantly hampers both the generalization ability and advancement of detection models.

3. Methodology

The proposed framework for face forgery detection is illustrated in Fig. 4. Firstly, we extract the fine-grained high-frequency component $x^{freq} \in \mathbb{R}^{h\times w\times 3}$ from the face image $x^{rgb} \in \mathbb{R}^{h\times w\times 3}$, where h and w represent the height and width of the input image, respectively. Subsequently, x^{rgb} and x^{freq} are fed into the Xception network to capture spatial and frequency features. The Feature Alignment Module (FAM) is employed in both shallow and deep layers of Xception to align and fuse the spatial-frequency features, enabling the exploration of their higher-order relationships. Finally, the Feature Independence Constrainer (FIC) eliminates statistical dependencies among the fused features to mitigate spurious correlations between irrelevant features and decision labels. We provide detailed descriptions of each module as well as implementation steps in the subsequent subsections.

3.1. Fine-grained high-frequency extraction

Joint frequency information has become a prevalent tool in the field of forgery detection. High frequencies capture the texture, contours, and artifacts of face images, while low frequencies typically represent semantic and background information. By emphasizing high frequencies and suppressing low frequencies, we can prevent the detection model from learning irrelevant correlations between features and labels to a certain extent. In our approach, we employ a sliding window frequency transform to decompose the original face image into multiple frequency components, thereby unveiling subtle forgery traces concealed in the frequency space. Specifically, we convert the input face image $x^{rgb} \in \mathbb{R}^{h\times w\times 3}$ into the YCbCr color space, i.e., $x^{ycbcr} \in \mathbb{R}^{h\times w\times 3}$. Then, x^{ycbcr} is divided into 8×8 patches $p^k_{i,j} \in \mathbb{R}^{8\times 8}$ using an 8×8 sliding window technique (where k denotes the corresponding color channel). We apply discrete cosine transform (DCT) to each patch $p^k_{i,j}$ to obtain its spectrum $d^k_{i,j} \in \mathbb{R}^{8\times 8}$. The direct current (DC) component of each $d^k_{i,j}$ is set to 0 while retaining only high-frequency information. Finally, all $d^k_{i,j}$ values are rearranged along channels to form fine-grained high-frequency components $x^{freq} \in \mathbb{R}^{h\times w\times 3}$. This fine-grained high-frequency information enables the model to detect subtle forgery cues in the spatial domain that are difficult to discern.

3.2. Feature alignment module

We incorporate fine-grained high-frequency information to enhance the detector's ability to detect subtle forgery artifacts. Due to the misalignment and insufficient interaction between spatial and frequency features (low-order relationships), the extracted frequency features lack spatial correlation, making it challenging to obtain more discriminative forgery clues. To address this issue, we propose the Feature Alignment Module (FAM) that aligns and integrates spatial and frequency domain features, as illustrated in Fig. 5.

Channel Alignment. The process of channel alignment involves concatenating the spatial feature $F^{rgb} \in \mathbb{R}^{H\times W\times C}$ with the frequency feature $F^{freq} \in \mathbb{R}^{H\times W\times C}$. Here, H and W represent the height and width of the feature map, respectively, while C denotes the channel. Subsequently, we compress the spatial information using a specific formula:

$$I = f_{GAP}([F^{rgb}, F^{freq}]), \tag{1}$$

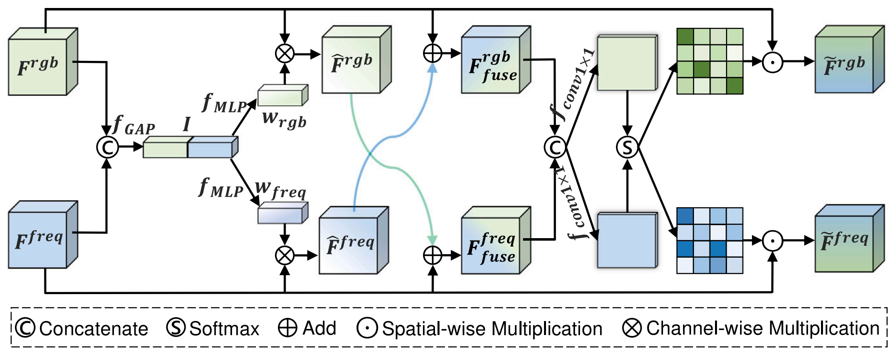

Fig. 5. The Feature Alignment Module (FAM) is designed to align and interact with spatial and frequency domain features to exploit their higher-order relationships.

where $f_{GAP}(\cdot)$ denotes the global average pooling, $[\cdot,\cdot]$ denotes concatenation operation, and $I = (I_1, \ldots, I_C, \ldots, I_{2C})$ is the global descriptor. To align F^{rgb} and F^{freq} on the channel, we introduce the multilayer perceptron (MLP) to compute the channel attention between them. The attention weights are then rescaled with a Sigmoid function:

$$w_{rgb} = Sigmoid(f_{MLP}(I)), \tag{2}$$

$$w_{freq} = Sigmoid(f_{MLP}(I)), \tag{3}$$

where $w_{rgb} \in \mathbb{R}^{1\times C}$ and $w_{freq} \in \mathbb{R}^{1\times C}$ denote the spatial and frequency channel attention weights, respectively. $Sigmoid(\cdot)$ denotes the Sigmoid function, and $f_{MLP}(\cdot)$ denotes the MLP layer. Then we can obtain the spatial feature $\hat{F}^{rgb}$ and frequency feature $\hat{F}^{freq}$ of the channel alignment:

$$\hat{F}^{rgb} = F^{rgb} \otimes w_{rgb}, \tag{4}$$

$$\hat{F}^{freq} = F^{freq} \otimes w_{freq}, \tag{5}$$

where $\otimes$ denotes channel-wise multiplication. To interact spatial and frequency features, we fuse and integrate them to exploit their higher-order relationships:

$$F^{rgb}_{fuse} = F^{rgb} + \hat{F}^{freq}, \tag{6}$$

$$F^{freq}_{fuse} = F^{freq} + \hat{F}^{rgb}. \tag{7}$$

Spatial Alignment. We spatially align spatial features and frequency features through positional attention. The formula is specified as:

$$\tilde{F}^{rgb} = F^{rgb} \odot \mathrm{Softmax}(f_{conv1\times1}(\left[F^{rgb}_{fuse}, F^{freq}_{fuse}\right])), \tag{8}$$

$$\tilde{F}^{freq} = F^{freq} \odot \mathrm{Softmax}(f_{conv1\times1}(\left[F^{rgb}_{fuse}, F^{freq}_{fuse}\right])), \tag{9}$$

where $f_{conv1\times1}(\cdot)$ is a 1×1 convolution, $\odot$ denotes spatial-wise multiplication. By aligning and interactively fusing spatial and frequency domain features, it allows mining higher-order relationships between them to discover subtle potential forgery traces. We place the FAM in the shallow and deep layers of Xception to align and fuse features.

3.3. Feature independence constrainer

The features extracted by the network contain detection-irrelevant information (e.g., face identity and content features), which significantly impacts the model's decision-making process. Particularly when there is a shift in the test set distribution, spurious correlations between these irrelevant features and decision labels can hinder model performance and generalization. Recent literature (Arjovsky, Bottou, Gulrajani, & Lopez-Paz, 2019; Lake, Ullman, Tenenbaum, & Gershman, 2017; Lopez-Paz, Nishihara, Chintala, Scholkopf, & Bottou, 2017; Marcus, 2018) has highlighted that these spurious correlations arise from the statistical dependency between relevant and irrelevant features. Building upon this concept, we propose Feature Independence Constrainer (FIC) to eliminate the statistical associations between features. For any pair of features X and Y, it is very hard to estimate the correlation between them. Inspired by the literature (Fukumizu, Gretton, Sun, & Schölkopf, 2007): the independence of variables X and Y can be equated to their cross-covariance operator being zero, i.e., $X \perp Y \Leftrightarrow \Sigma_{XY} = 0$. Therefore, according to the Hilbert–Schmidt Independence Criterion (Gretton et al., 2007), to eliminate the dependence between X and Y, it is sufficient to force their cross-covariance operator to converge to zero (Bahng, Chun, Yun, Choo, & Oh, 2020). We can obtain $\Sigma_{XY} = E_{XY}[a(X)b(Y)] - E_X[a(X)]E_Y[b(Y)]$ following the literature (Fukumizu, Bach, & Jordan, 2004), where $E[\cdot]$ is the expectation and $a(\cdot)$ and $b(\cdot)$ are the mapping functions on the Reproducing Kernel Hilbert Space. Since $a(\cdot)$ and $b(\cdot)$ are implicit, we estimate the kernel function by random Fourier equations (Rahimi & Recht, 2007). The covariance matrix can be converted into the following form:

$$\begin{aligned}\Sigma_{XY} &= \begin{bmatrix} \sigma(X_1, Y_1) & \cdots & \sigma(X_1, Y_n) \\ \vdots & \ddots & \vdots \\ \sigma(X_n, Y_1) & \cdots & \sigma(X_n, Y_n) \end{bmatrix} \\ &= \frac{1}{n-1}\sum_{d=1}^{n}[(g(X_d) - E[g(X)])^T \cdot (h(Y_d) - E[h(Y)])],\end{aligned} \tag{10}$$

where $E[g(X)] = \frac{1}{n}\sum_{m=1}^{n} g(X_m)$, $E[h(Y_m)] = \frac{1}{n}\sum_{m=1}^{n} h(Y_m)$, n is the number of samples in a batch. $(X_1, \ldots, X_n)$ and $(Y_1, \ldots, X_n)$ are obtained by sampling from the distributions of X and Y, respectively. $g(\cdot)$ and $h(\cdot)$ are sampled from the space of random Fourier equations, i.e., $g(x), h(x) = \sqrt{\frac{2}{D}}[\cos(\omega_1^T x + \rho_1), \ldots, \cos(\omega_D^T x + \rho_D)]$, where $(\omega_1, \ldots, \omega_D)$ is sampled from the standard normal distribution, and $(\rho_1, \ldots, \rho_D)$ is sampled from the uniform distribution $[0, 2\pi]$. In our experiments we set D to 5. Since Σ_{XY} is invariant, to allow Σ_{XY} to dynamically decrease gradually as the model iterates, we assemble a set of dynamic weight parameter $c = (c_1, \ldots, c_n)$ for Σ_{XY}:

$$\Sigma_{XY,c} = \frac{1}{n-1}\sum_{d=1}^{n}[(c_d g(X_d) - E[cg(X)])^T \cdot (c_d h(Y_d) - E[ch(Y)])], \tag{11}$$

where $E[cg(X)] = \frac{1}{n}\sum_{m=1}^{n} c_m g(X_m)$, $E[ch(Y)] = \frac{1}{n}\sum_{m=1}^{n} c_m h(Y_m)$, $\sum_{m=1}^{n} c_m = n$. We measure the correlation between features utilizing the Frobenius norm (Strobl, Zhang, & Visweswaran, 2019), i.e., $\left\|\Sigma_{XY,c}\right\|_F^2$. Thus, the statistical dependence between features is eliminated by optimizing the dynamic weight parameter c.

$$c' = \arg\min_{c} \sum_{i=1}^{M}\sum_{j=1}^{M}\left\|\Sigma_{X_i Y_j, c}\right\|_F^2, \tag{12}$$

where M denotes the dimensions of the features X and Y. To facilitate optimization, the dynamic weight parameter c is applied to the final classification loss to balance the distribution of each feature variable.

$$f'_{det} = \arg\min_{f_{det}} \sum_{i=1}^{n} c_i \mathcal{L}(I^i_{pre}, I^i_{lab}), \tag{13}$$

where f_{det} is the detection network, I^i_{pre} and I^i_{lab} are the network's prediction for the ith image and the corresponding label, respectively.

$\mathcal{L}(\cdot)$ is the cross-entropy loss function. In this way, the statistical associations between relevant and irrelevant features are eliminated, which mitigates spurious correlations, and in turn the model can be generalized to detect unknown forgeries.

4. Experiments and performance analysis

In this section, we present the evaluation and analysis of the performance of the proposed method through a series of experiments. The experiments consist of five parts. First, we provide an overview of the datasets and their pre-processing procedures. Second, we describe the specific experimental setup employed. Third, we assess the detection performance of the proposed method through both numerical and qualitative results, including intra-dataset, cross-method, and cross-dataset comparisons. Fourth, we evaluate the robustness of the proposed method against various common content-preserving manipulations (such as Gaussian noise, Gaussian blur, JPEG compression) as well as unknown perturbations using quantitative indicators. Lastly, ablation experiments are conducted to validate the significance of each module. All these performance metrics are thoroughly analyzed and compared with state-of-the-art methods.

4.1. Datasets

To evaluate the generalization and robustness of our method, we conduct extensive qualitative and quantitative experiments. We set up four comprehensive experiments: intra-dataset experiments, cross-method experiments (with unseen manipulations), cross-dataset experiments (with unseen datasets), and experiments with uncertain perturbations.

(1) Intra-dataset experiments. The widely used dataset FaceForensics++ (FF++) (Rossler et al., 2019) is employed for intra-dataset evaluation, consistent with previous studies (Chen, Lin, Li, & Tan, 2022; Zhao et al., 2023). FF++ comprises 1000 original videos sourced from YouTube and 4000 manipulated videos generated through four distinct forgery techniques: FaceSwap (FS), Face2Face (F2F), Deepfakes (DF), and NeuralTextures (NT). Adhering strictly to the official dataset partitioning criteria, we allocate 720 videos for training, 140 for validation, and 140 for testing. To ensure comprehensive training and evaluation, we select two datasets with varying compression ratios: C23 for high-quality videos and C40 for low-quality ones.

(2) Cross-method experiments. To verify the performance of the proposed method on unknown types of forgery, we train on one forgery type from FF++ (C23) and test on the other three. Specifically, we train on Deepfakes (DF) and test on FaceSwap (FS), Face2Face (F2F), and NeuralTextures (NT). The remaining three combinations follow a similar approach.

(3) Cross-dataset experiments. To evaluate the performance of our method on unseen forged datasets, we have selected datasets that have undergone extensive testing to assess their generalizability. These datasets are described as follows: (1) Celeb-DF (Li, Yang, Sun, Qi and Lyu, 2020) consists of two versions: CDF(V1) and CDF(V2). CDF(V1) includes 408 real videos and 795 fake videos, while CDF(V2) contains 5639 high-quality fake videos generated from 590 real videos. (2) DeepfakeTIMIT (TIMIT) (Korshunov & Marcel, 2018) has two quality versions: TIMIT(HQ) and TIMIT(LQ), with 320 high-quality and 320 low-quality fake videos, respectively. (3) Deep Fake Detection (DFD) (Nick & Andrew, 2019) comprises 363 real videos and 3068 fake videos. (4) UADFV (Yang, Li, & Lyu, 2019) consists of low-quality real and fake videos, each category containing a total of 49 samples.

(4) Adding uncertain perturbations experiments. We evaluate the impact of perturbations on the performance of our approach by conducting experiments on the DeeperForensics-1.0 (Jiang, Li, Wu, Qian, & Loy, 2020) dataset, which comprises 48,475 source videos and 11,000 manipulated videos with seven types of real-world perturbations at five intensity levels. Specifically, we use 1000 manipulated videos from the standard set (std) for training and another 3000 for testing. The test set includes: 1000 videos with a single level of random distortion (std/sing), 1000 with a random level of distortion (std/rand), and 1000 with a mixture of three random levels and kinds of distortions (std/mix).

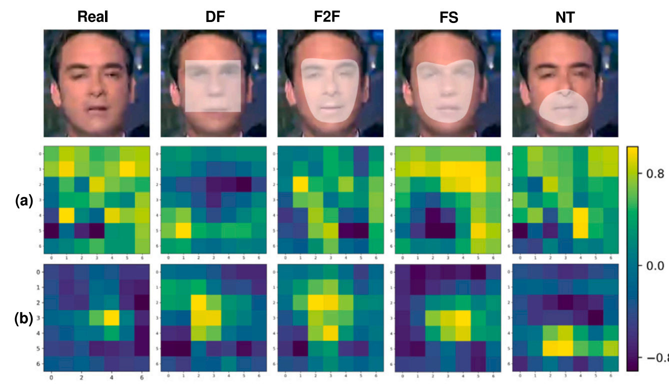

Fig. 6. Feature visualization for four kinds of forged faces on FF++ (C40). Rows (a) and (b) correspond to the feature maps learned by the last convolutional layer of Xception and ours, respectively.

4.2. Experimental setup

For data preprocessing, we intercept 32 frames at equal intervals in each video and resize the facial images to 224 × 224. In the training phase, we set the batch size to 64. We use Adam (Kingma & Ba, 2014) as the optimizer of our model with an initial learning rate of 0.001 and decay by half every 5 epochs. The training epochs is 30. We use Xception as the backbone network. We employ ACC (Accuracy score) and AUC (Area Under the Receiver Operating Characteristic Curve) (frame-level) as evaluation metrics. All codes are based on the PyTorch and trained on an NVIDIA RTX 3090 GPU.

4.3. Comparison with state-of-the-art methods

In this subsection, we compare our method with current state-of-the-art (SOTA) deepfake detection methods. We evaluate the performance of our method on the intra-dataset scenario. Additionally, we test its generalization capabilities, including cross-method and cross-dataset evaluations.

(1) Intra-dataset Evaluation. To comprehensively evaluate the detection performance of our method, we conducted a comparative analysis with 17 state-of-the-art (SOTA) deepfake detection methods on the FF++ (C23, C40) dataset. The results are presented in Table 1. All approaches were trained and tested using images of identical quality, specifically training on FF++ (C23) and testing on FF++ (C23). From Table 1, it can be observed that our method demonstrates comparable performance on C23. However, when compared to some SOTA methods, our model shows a slight decrease in accuracy for C23. This discrepancy can be attributed to the fact that most SOTA methods utilize diverse data augmentations to enhance detection accuracy, which is not the case with our approach. It is worth noting that face detection methods generally face challenges when dealing with degraded image quality. Analyzing experiments conducted on C40 (low quality), our method continues to exhibit strong performance in handling low-quality face images while significantly outperforming other techniques.

To provide a more intuitive demonstration of the effectiveness of our proposed method, we conducted a qualitative evaluation of the

Table 1
Performance (%) comparison with 17 state-of-the-art methods on the FF++ (C23, C40) dataset. The best and second best results are marked in **Bold** and <u>Underline</u>, respectively.

Methods	Publication	FF++ (C23)		FF++ (C40)	
		ACC	AUC	ACC	AUC
MesoNet (Afchar, Nozick, Yamagishi, & Echizen, 2018)	WIFS'17	83.10	84.30	70.47	72.62
Multi-task (Nguyen, Fang, Yamagishi and Echizen, 2019)	BTAS'19	85.65	85.43	81.30	75.59
Xception (Chollet, 2017)	ICCV'19	95.73	96.30	86.86	89.30
Face X-ray (Li, Bao, Zhang et al., 2020)	CVPR'20	–	87.40	–	61.60
F3-Net (Qian et al., 2020)	ECCV'20	**97.52**	98.10	90.43	93.30
Two-branch (Masi, Killekar, Mascarenhas, Gurudatt, & AbdAlmageed, 2020)	ECCV'20	96.43	98.70	86.34	86.59
Add-Net (Zi, Chang, Chen, Ma, & Jiang, 2020)	ACMMM'20	96.78	97.74	87.50	91.01
Multi-Attention (Zhao et al., 2021)	CVPR'21	96.37	98.97	86.95	87.26
Freq-SCL (Li, Xie, Li, Wang, & Zhang, 2021)	CVPR'21	96.69	99.28	89.00	92.39
RFM (Wang & Deng, 2021)	CVPR'21	95.69	98.79	87.06	89.83
SPSL (Liu et al., 2021)	CVPR'21	91.50	95.32	81.57	82.82
RECCE (Cao et al., 2022)	CVPR'22	97.06	**99.32**	91.03	95.02
HFI-Net (Miao, Tan, Chu, Yu, & Guo, 2022)	TIFS'22	95.12	98.66	85.69	88.40
LiSiam (Wang, Sun and Tang, 2022)	TIFS'22	96.51	99.13	87.81	91.44
MRL (Yang, Liang et al., 2023)	TIFS'23	93.82	98.27	<u>91.81</u>	<u>96.18</u>
SFIConv (Guo, Jia et al., 2023)	TIFS'24	93.61	98.46	89.41	95.90
ID3 (Yin et al., 2024)	TMM'24	93.76	97.82	83.64	86.38
Ours		<u>97.14</u>	<u>99.29</u>	**92.30**	**96.74**

Table 2
Cross-method evaluation in terms of AUC (%) on different manipulation techniques. Gray background means within-dataset results. The best and second best results are marked in **Bold** and <u>Underline</u>, respectively.

Methods	Train set	Test set				Cross Ave.
		DF	FS	F2F	NT	
Freq-SCL (Li et al., 2021)	DF	98.91	66.87	58.90	63.61	63.13
Multi-Attention (Zhao et al., 2021)	DF	99.51	67.33	66.41	66.01	65.58
RECCE (Yang, Li, Xiao, Lu, & Gao, 2021)	DF	<u>99.65</u>	**74.29**	70.66	67.34	70.76
DCL (Sun, Yao et al., 2022)	DF	**99.98**	61.06	<u>77.13</u>	**75.01**	<u>71.07</u>
Ours	DF	99.47	<u>69.06</u>	**77.39**	<u>68.51</u>	**71.65**
Freq-SCL (Li et al., 2021)	FS	75.90	98.37	54.64	49.72	60.09
Multi-Attention (Zhao et al., 2021)	FS	<u>82.33</u>	98.82	61.65	54.79	66.26
RECCE (Yang et al., 2021)	FS	**82.39**	98.82	64.44	<u>56.70</u>	<u>67.84</u>
DCL (Sun, Yao et al., 2022)	FS	74.80	**99.90**	**69.75**	52.60	65.72
Ours	FS	81.47	<u>98.93</u>	<u>65.28</u>	**60.63**	**69.13**
Freq-SCL (Li et al., 2021)	F2F	67.55	55.35	93.06	66.66	63.19
Multi-Attention (Zhao et al., 2021)	F2F	73.04	<u>65.10</u>	97.96	71.88	70.01
RECCE (Yang et al., 2021)	F2F	75.99	64.53	98.06	<u>72.32</u>	70.95
DCL (Sun, Yao et al., 2022)	F2F	**91.91**	59.58	**99.21**	66.67	<u>72.72</u>
Ours	F2F	<u>78.07</u>	**67.58**	<u>98.27</u>	**74.01**	**73.22**
Freq-SCL (Li et al., 2021)	NT	79.09	53.99	74.21	88.54	69.10
Multi-Attention (Zhao et al., 2021)	NT	74.56	60.90	<u>80.61</u>	93.34	72.02
RECCE (Yang et al., 2021)	NT	78.83	63.70	**80.89**	<u>93.63</u>	<u>74.47</u>
DCL (Sun, Yao et al., 2022)	NT	**91.23**	**79.31**	52.13	**98.97**	74.22
Ours	NT	<u>83.81</u>	<u>63.88</u>	78.60	92.42	**75.43**

feature maps learned by the last convolutional layer of both our method and the baseline model on C40. As depicted in Fig. 6, it is evident that Xception captures numerous irrelevant details, such as background information, which are unrelated to forgery features. In contrast, our approach effectively avoids interference from these irrelevant features and focuses primarily on artifacts. Specifically, even for local forgery techniques like NeuralTextures (NT), such as lip movement, our model successfully learns the corresponding forgery features while disregarding irrelevant ones. Notably, the feature visualization of our model outperforms Xception even on the highly compressed FF++ (C40) dataset.

(2) Cross-method Evaluation. We also conducted cross-method experiments on FF++ (C23), where the model was trained on one type of forgery and tested on the other three types. In Table 2, we compare our results with recent state-of-the-art methods. Our approach shows significant superiority over the frequency domain-based method Freq-SCL (Li et al., 2021), even though Freq-SCL incorporates an auxiliary loss function. While DCL (Sun, Yao et al., 2022) achieves strong performance through contrast learning strategies, our approach does not rely on additional data enhancement or learning strategies but still exhibits better generalization across different methods compared to existing state-of-the-art approaches. As shown in Fig. 7, spurious correlation severely affects the generalization of the detection model Xception. Thanks to FIC's ability to eliminate statistical dependencies between features and thus mitigate spurious correlations, the combination of Xception and FIC significantly improves performance in cross-method scenarios. Additionally, we observe that incorporating fine-grained high-frequency information helps mitigate the influence of spurious correlations between irrelevant features and labels.

(3) Cross-dataset Evaluation. To evaluate the generalization capability of our method on unseen forgeries, we conduct cross-dataset experiments by training and testing on distinct datasets. Specifically, we train the model separately on FF++ (C23) and FF++ (C40), then test it on CDF(V1), CDF(V2), TIMIT(HQ), TIMIT(LQ), UADFV, and DFD. The discrepancy between the test dataset and training data presents a more challenging scenario. The results are presented in Table 3. Previous state-of-the-art methods, such as F3-Net (Qian et al., 2020) and Multi-Attention (Zhao et al., 2021), excel in intra-class testing but exhibit significantly degraded performance when faced with unseen datasets like CDF(V1) and CDF(V2). These methods struggle with generalization

Table 3
Cross-dataset evaluation in terms of AUC (%) on unseen datasets. The best and second best results are marked in **Bold** and Underline, respectively (Dang, Liu, Stehouwer, Liu, & Jain, 2020; Fei et al., 2022; Miao et al., 2021; Nguyen, Yamagishi and Echizen, 2019).

Methods	Train set	Unseen test set					
		CDF(V1)	CDF(V2)	TIMIT(LQ)	TIMIT(HQ)	UADFV	DFD
Multi-task (Nguyen, Yamagishi et al., 2019)[a]	FF++	36.50	54.30	62.20	53.60	65.80	–
Capsule (Jiang et al., 2020)[a]	FF++	–	57.50	78.40	74.40	61.30	–
FFD (Dang et al., 2020)[b]	DFFD	64.40	–	–	–	–	–
LDA (Lu et al., 2023)[a]	FF++	–	70.33	–	–	–	–
Xception (Chollet, 2017)	C23	64.16	65.01	88.35	90.73	69.50	70.81
F3-Net (Qian et al., 2020)	C23	63.70	65.28	86.41	89.05	73.78	76.04
Multi-Attention (Zhao et al., 2021)	C23	62.04	67.27	81.39	87.77	72.96	81.52
LiSiam (Wang, Sun et al., 2022)	C23	**81.14**	<u>78.21</u>	–	–	–	–
SOLA (Fei et al., 2022)	C23	–	76.02	–	–	–	–
Xception+FIC	C23	75.62	73.08	<u>91.47</u>	<u>93.90</u>	<u>75.85</u>	<u>86.29</u>
Ours	C23	<u>79.85</u>	**79.36**	**93.37**	**95.09**	**81.40**	**90.50**
Xception (Chollet, 2017)	C40	62.87	64.31	74.69	72.50	50.71	63.43
F3-Net (Qian et al., 2020)	C40	65.74	66.90	<u>78.53</u>	<u>82.62</u>	**72.68**	71.55
Multi-Attention (Zhao et al., 2021)	C40	64.30	62.85	75.39	77.40	67.19	76.28
SPSL (Liu et al., 2021)	C40	<u>76.88</u>	–	–	–	–	–
FRLM (Miao et al., 2021)	C40	76.52	70.58	70.68	65.48	64.20	–
Xception+FIC	C40	69.10	<u>71.48</u>	77.03	76.40	63.53	<u>78.85</u>
Ours	C40	**77.53**	**78.60**	**81.74**	**83.05**	<u>71.31</u>	**83.72**

[a] Indicates that compression ratio of the training set is undocumented.
[b] Indicates that additional datasets are employed for training.

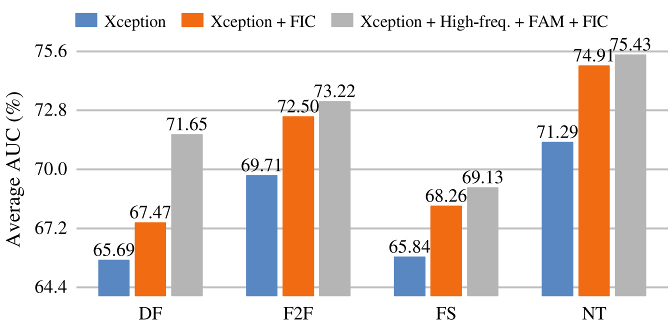

Fig. 7. Performance improvements of fine-grained high-frequency (High-freq.), FIC and FAM for the baseline model (i.e., Xception). The models are trained on one kind of data from FF++ dataset (C23) and tested on the other three kinds, e.g., trained on Deepfake (DF), tested on Face2Face (F2F), FaceSwap (FS) and NeuralTextures (NT). The horizontal and vertical coordinates indicate the training set and the average AUC, respectively.

Table 4
Performance (ACC (%)) comparison with the SOTAs on DeeperForensics-1.0 dataset. The training set is Std.

Methods	Test		
	std/rand	std/sing	std/mix
Xception (Chollet, 2017)	93.28	85.14	78.75
M2TR (Wang, Wu et al., 2022)	94.78	**91.79**	86.32
MTD-Net (Yang et al., 2021)	95.22	–	86.89
F3-Net (Qian et al., 2020)	94.06	87.79	81.23
Multi-Attention (Zhao et al., 2021)	92.75	84.91	84.26
Ours	**96.24**	88.07	**89.64**

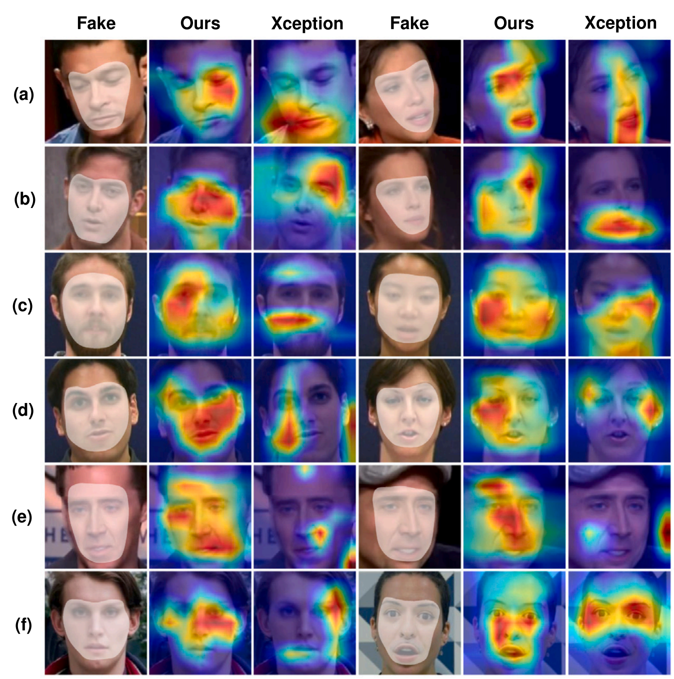

Fig. 8. Grad-CAM visualization of the feature maps on cross-dataset. Training on FF++ (C23), testing on unseen datasets. (a) CDF(V1). (b) CDF(V2). (c) TIMIT(LQ). (d) TIMIT(HQ). (e) UADFV. (f) DFD. The 2nd/5th and 3rd/6th columns for the proposed method and Xception, respectively.

due to spurious correlation issues. In comparison, our approach outperforms existing techniques in terms of generalization across multiple datasets. Notably, training on the low-quality dataset FF++ (C40) and testing on CDF(V1) and TIMIT(LQ) is particularly demanding. Our approach achieves an AUC of 77.53% and 81.74%, surpassing F3-Net by 11.79% and 3.21%, respectively. Table 3 shows that integrating FIC significantly enhances the generalization performance of the Xception detection model across diverse datasets, effectively mitigating spurious correlations. As shown in Fig. 8, we compare the proposed method with a baseline model (Xception) using Grad-CAM to visualize the internal feature representations learned by the network. While Xception predominantly focuses on background and local semantic features, our model prioritizes forgery features due to its ability to reduce reliance on irrelevant features, ensuring successful generalization to unseen datasets.

4.4. Robustness evaluation

The utilization of image processing is prevalent in social media, and the robustness of the model against uncertain perturbations plays a crucial role. We conduct a comprehensive evaluation of robustness

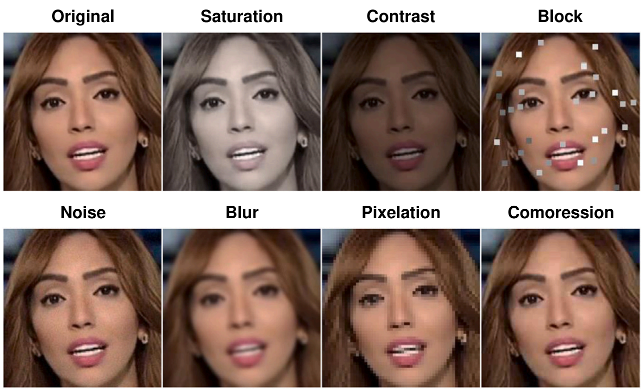

Fig. 9. Corruption examples on FF++ (C23). It consists of saturation change, contrast change, adding distortion blocks, Gaussian noise, Gaussian blur, pixelation, and JEPG compression.

on the DeeperForensics-1.0 dataset, which encompasses diverse levels and types of distortions. The models are trained on the standard set (std) and tested on three perturbation sets with distinct configurations (i.e., std/rand, std/sing, std/mix). Our method exhibits superior stability against unknown perturbations compared to the state-of-the-art methods listed in Table 4. Notably, M2TR (Wang, Wu et al., 2022) and F3-Net (Qian et al., 2020) demonstrate relatively stable performance when confronted with uncertain perturbations. This observation suggests that relying solely on spatial domain feature extraction is constrained by the presence of perturbations.

In addition, we train the models on FF++ (C23) and subsequently evaluate their robustness by testing them on various corrupted versions of FF++ (C23). We consider seven types of erosion operations: saturation change, contrast change, distortion block addition, Gaussian noise, Gaussian blur, pixelation, and JPEG compression. Each operation consists of five levels ranging from light to heavy. Fig. 9 provides examples for each level of erosion. We compare our method with three state-of-the-art approaches (namely Xception, F3-Net (Qian et al., 2020), and Multi-Attention (Zhao et al., 2021)), where F3-Net is a frequency domain-based method while the other two are spatial-based methods. As depicted in Fig. 10, our method exhibits greater stability against most perturbations compared to these state-of-the-art methods. Notably, our method maintains stable performance for image addition distortion blocks and Gaussian blur. Moreover, even under light JPEG compression conditions, our method achieves an impressive 80% AUC performance, surpassing that of the other baseline methods. However, all methods essentially fail when subjected to heavy JPEG compression. Additionally, our method shows sensitivity to Gaussian noise due to its extraction of excessive high-frequency information, which can be disruptive. Consequently, the model struggles to effectively learn discriminative features. This observation is further supported by the significant performance degradation exhibited by F3-Net, which relies solely on frequency information and suffers from severe deterioration in the presence of noise. It should be noted that images with a JPEG compression quality factor below 20 experience substantial degradation and distortion, characterized by jagged edges and noticeable color-block artifacts. The detection of such extremely low-quality face images holds little practical significance.

4.5. Ablation studies

To evaluate the impact of each component on our approach, we conducted ablation experiments on fine-grained high-frequency (High-freq.), FIC, and FAM. The model was trained on FF++ (C40), and its performance was evaluated on both FF++ (C40) and CDF(V1). The results are presented in Table 5. Compared to the baseline (Xception), our model achieved an improvement of up to 7.44% in AUC for C40 and 12.66% for CDF(V1). However, removing FIC resulted in a decrease of 8.09% in generalization performance. This demonstrates that FIC effectively mitigates spurious correlation, thereby enhancing the model's generalizability. Additionally, by exploiting higher-order correlations between features in the spatial and frequency domains, FAM enhances the detection accuracy and cross-domain capability of the model. In Fig. 11, we visualize the feature map of the last convolutional layer of our model for all four forgery types present in the C40 dataset. The fifth column shows the higher response to manipulated regions for all forgery types. The absence of FIC leads the model to learn many features unrelated to artifacts and indicates that FIC enables the model to focus more on the features of the forged region while ignoring the interference from irrelevant features.

Table 5
Experimental results (AUC (%)) for the effect of different components of our approach. Each model is trained by FF++ (C40) and tested on FF++ (C40), and CDF(V1). The proposed method shows a significant improvement in cross-dataset evaluation.

Model	High-freq.	FAM	FIC	FF++ (C40)	CDF(V1)
	✗	✗	✗	89.30	62.87
	✗	✗	✓	91.75 (↑2.45)	69.10 (↑6.23)
	✓	✗	✗	93.84 (↑4.54)	64.28 (↑1.41)
Xception	✓	✗	✓	95.60 (↑6.30)	74.29 (↑11.42)
	✓	✓	✗	95.19 (↑5.89)	67.44 (↑4.57)
	✓	✓	✓	**96.74** (↑7.44)	**77.53** (↑14.66)

Table 6
Ablation (AUC (%)) of HFE position. Block3 and Block11 indicate the shallow and deep Blocks of Xception, respectively.

Block3	Block11	FF++ (C40)	CDF(V1)
✗	✗	95.60	74.29
✓	✗	95.80 (↑0.20)	76.16 (↑1.87)
✗	✓	96.25 (↑0.65)	75.30 (↑1.01)
✓	✓	**96.74** (↑1.14)	**77.53** (↑3.24)

The position of FAM was also ablated in Table 6. Our findings demonstrate that incorporating simultaneous fusion and alignment of spatial and frequency domain features at both shallow (Block3) and deep (Block11) layers significantly enhances detection performance and overall generalization capability of the model.

5. Conclusion and future work

In this study, we analyze the main reasons for detector cross-domain failures from a causal learning perspective and identify that the essential problem lies in the spurious correlation between irrelevant features and category labels. To address this issue, we propose the Feature Independence Constrainer (FIC), which aims to mitigate spurious correlations and improve the model's generalization performance. Specifically, FIC maps features to a Reproducing Hilbert Space and computes the covariance matrix between kernel-mapped features. FIC achieves iterative optimization by assembling the covariance matrix with dynamic weight parameters. This approach eliminates statistical dependencies between features and ensures that each pair of features remains independent, thereby reducing the impact of spurious correlations. Additionally, during the data preprocessing stage, we inject fine-grained high-frequency components into the backbone network to enable the detection model to better learn artifactual concepts related to forgery and to avoid some degree of spurious correlations between irrelevant features and labels. We also design the Feature Alignment Module (FAM) to explore higher-order correlations between the spatial and frequency domains and to mine richer potential forgery traces. Quantitative and qualitative experiments demonstrate the superiority of the proposed method over state-of-the-art approaches on multiple face forgery benchmark datasets.

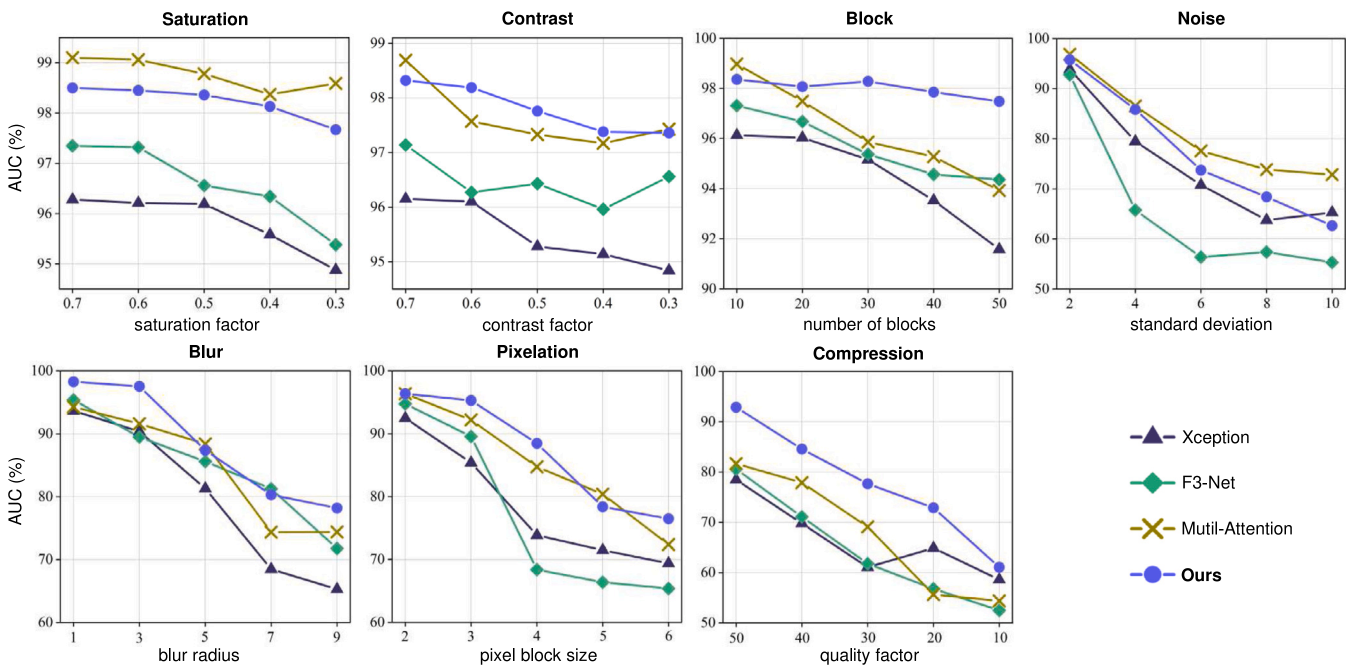

Fig. 10. Robustness evaluation for 7 kinds of perturbations. 5 levels are set for each kind of disturbance.

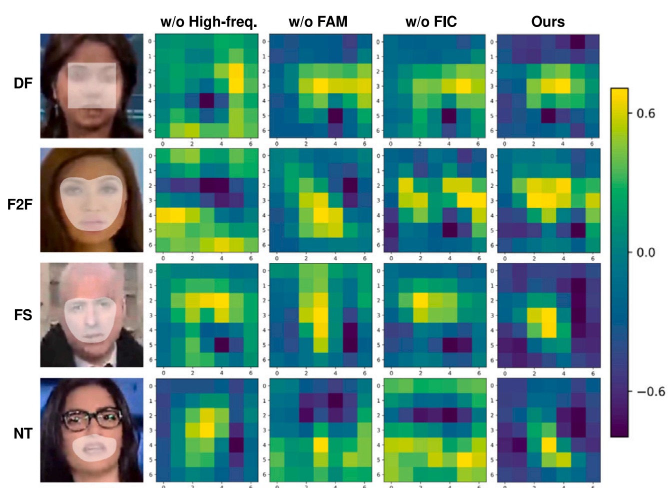

Fig. 11. Feature visualizations for four kinds of manipulated faces on FF++ (C40). The first column indicates the manipulated faces, and the rest columns are feature maps corresponding to 'w/o High-freq', 'w/o FAM', 'w/o FIC' and ours, respectively. 'w/o' denotes 'without'.

Our method also has certain limitations. We are unable to quantitatively measure the spurious correlation of each feature extracted by the detection model. Accurately quantifying the spurious purity of features would enhance the interpretability of model decisions. We will strive to address these issues in future research to improve the reliability of detection models.

CRediT authorship contribution statement

Ningning Bai: Writing – original draft, Validation, Methodology. **Xiaofeng Wang:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis. **Ruidong Han:** Investigation, Data curation, Conceptualization. **Jianpeng Hou:** Validation, Software. **Qin Wang:** Writing – review & editing, Investigation, Data curation. **Shanmin Pang:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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